# Part 1: Environment Setup & Mini Project

## Group Members


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# 1. Framework Choice & Installation

## Which Framework Was Selected

**Framework:** TensorFlow and PyTorch both

## Installation Method

**Method Used:** pip

**Commands Executed:**

```
pip install tensorflow
pip install torch torchvision
```

**Installation Status:** ✅ Successful

```
(myenv) muditha@lap:~/Applied-artificial-intelligence$ python3 --version
Python 3.10.12
(myenv) muditha@lap:~/Applied-artificial-intelligence$ python -c "import tensorflow as tf; print(tf.__version__)"
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1776207136.647845   45154 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point r
ound-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
I0000 00:00:1776207136.649956   45154 cudart_stub.cc:31] Could not find cuda drivers on your machine, GPU will not be used.
I0000 00:00:1776207136.926408   45154 cpu_feature_guard.cc:227] This TensorFlow binary is optimized to use available CPU instructions in performance-critica
l operations.
To enable the following instructions: AVX2 AVX512F AVX512_VNNI FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1776207138.326102   45154 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point r
ound-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
I0000 00:00:1776207138.327748   45154 cudart_stub.cc:31] Could not find cuda drivers on your machine, GPU will not be used.
2.21.0
(myenv) muditha@lap:~/Applied-artificial-intelligence$ python -c "import torch; print(torch.__version__)"
2.11.0+cu130
```

# 2. Virtual Environment Setup

## Environment Configuration

**Virtual Environment Tool:** venv

**Setup Commands:**

```
python3 -m venv myenv
source myenv/bin/activate
```

```
(myenv) muditha@lap:~/Applied-artificial-intelligence$
```

## Environment Details

- **Python Version:** 3.10.12
- **Framework Version:** TensorFlow 2.21.0, PyTorch 2.11.0+cu130


```
(myenv) muditha@lap:~/Applied-artificial-intelligence$ python3 --version
Python 3.10.12
(myenv) muditha@lap:~/Applied-artificial-intelligence$ python -c "import tensorflow as tf; print(tf.__version__)"
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WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1776207138.326102   45154 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point r
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I0000 00:00:1776207138.327748   45154 cudart_stub.cc:31] Could not find cuda drivers on your machine, GPU will not be used.
2.21.0
(myenv) muditha@lap:~/Applied-artificial-intelligence$ python -c "import torch; print(torch.__version__)"
2.11.0+cu130
```


- **Dependencies Installed:**

```
(myenv) muditha@lap:~/Applied-artificial-intelligence$ pip list
Package                  Version
------------------------ ---------
absl-py                  2.4.0
astunparse               1.6.3
certifi                  2026.2.25
charset-normalizer       3.4.7
cuda-bindings            13.2.0
cuda-pathfinder          1.5.3
cuda-toolkit             13.0.2
filelock                 3.25.2
flatbuffers              25.12.19
fsspec                   2026.3.0
gast                     0.7.0
google-pasta             0.2.0
grpcio                   1.80.0
h5py                     3.14.0
idna                     3.11
Jinja2                   3.1.6
keras                    3.12.1
libclang                 18.1.1
markdown-it-py           4.0.0
MarkupSafe               3.0.3
mdurl                    0.1.2
ml_dtypes                0.5.4
mpmath                   1.3.0
namex                    0.1.0
networkx                 3.4.2
numpy                    2.2.6
nvidia-cublas            13.1.0.3
nvidia-cuda-cupti        13.0.85
nvidia-cuda-nvrtc        13.0.88
nvidia-cuda-runtime      13.0.96
nvidia-cudnn-cu13        9.19.0.56
nvidia-cufft             12.0.0.61
nvidia-cufile            1.15.1.6
nvidia-curand            10.4.0.35
nvidia-cusolver          12.0.4.66
nvidia-cusparse          12.6.3.3
nvidia-cusparselt-cu13 0.8.0
nvidia-nccl-cu13         2.28.9
nvidia-nvjitlink         13.0.88
nvidia-nvshmem-cu13      3.4.5
nvidia-nvtx              13.0.85
opt_einsum               3.4.0
optree                   0.19.0
packaging                26.1
```

```
pillow                  12.2.0
pip                     26.0.1
protobuf                7.34.1
Pygments                2.20.0
requests                2.33.1
rich                    15.0.0
setuptools              59.6.0
six                     1.17.0
sympy                   1.14.0
tensorflow              2.21.0
termcolor               3.3.0
torch                   2.11.0
torchaudio              2.11.0
torchvision             0.26.0
triton                  3.6.0
typing_extensions       4.15.0
urllib3                 2.6.3
wheel                   0.46.3
wrapt                   2.1.2
(myenv) muditha@lap:~/Applied-artificial-intelligence$
```

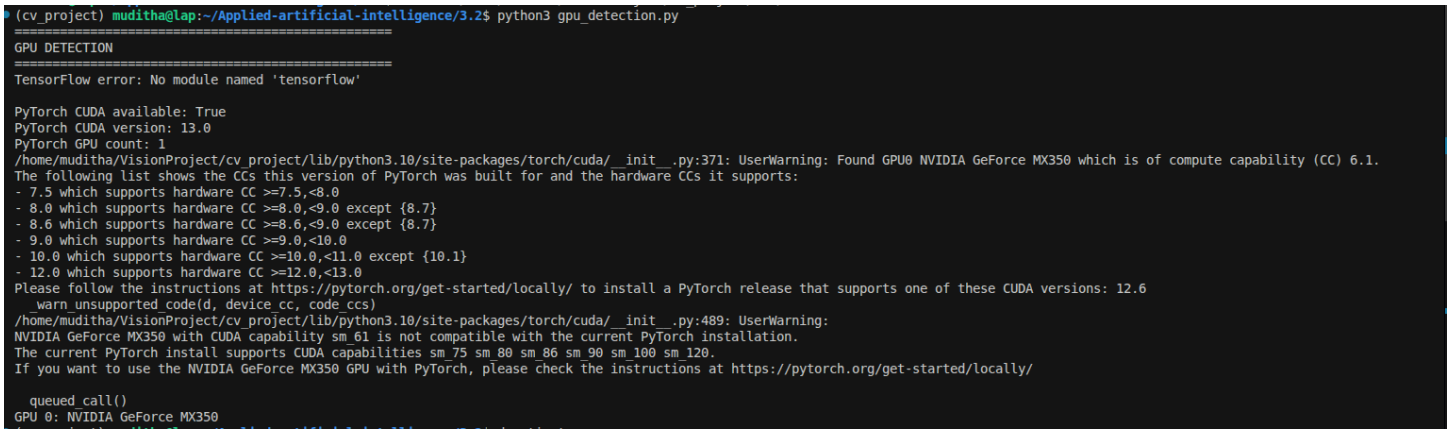
# 3. GPU Verification & Configuration

## GPU Detection Code

```
"""GPU detection script for TensorFlow and PyTorch."""

def check_tensorflow_gpu():
    try:
        import tensorflow as tf

        gpus = tf.config.list_physical_devices('GPU')
        print('TensorFlow GPUs:', gpus)
        print('TensorFlow GPU count:', len(gpus))
    except Exception as e:
        print('TensorFlow error:', e)


def check_pytorch_gpu():
    try:
        import torch

        print('PyTorch CUDA available:', torch.cuda.is_available())
        print('PyTorch CUDA version:', torch.version.cuda)
        print('PyTorch GPU count:', torch.cuda.device_count())

        if torch.cuda.is_available():
            for i in range(torch.cuda.device_count()):
                print(f'GPU {i}:', torch.cuda.get_device_name(i))
    except Exception as e:
        print('PyTorch error:', e)


if __name__ == '__main__':
    print('=' * 50)
    print('GPU DETECTION')
    print('=' * 50)
```

```
check_tensorflow_gpu()
print()
check_pytorch_gpu()
```

**Output Screenshot:**




# GPU Status

- **GPU Available:** ✅ Yes
- **Number of GPUs Detected:** 1
- **GPU Name(s):** NVIDIA GeForce MX350
- **CUDA Version:** 13.0 (PyTorch build)
- **Compatibility Note:** ⚠️ GPU compute capability is sm_61, but the installed PyTorch build supports sm_75 and above.

# Any Issues Encountered & Resolution

- **Issue 1 (PyTorch GPU compatibility):** PyTorch detected the MX350 GPU, but raised warnings that CUDA capability sm_61 is not compatible with the installed build.
  - **Resolution:** Run PyTorch on CPU in the current setup. Because my GPU is very old.

# 4. Minimal Test Script

## Test Code Implementation

### TensorFlow Example

```
import tensorflow as tf
import numpy as np
import os

# Force TensorFlow to use CPU only due to older VGA
os.environ["CUDA_VISIBLE_DEVICES"] = "-1"

# Verify environment isolation and device usage
print("TensorFlow version:", tf.__version__)
print("Available devices:", tf.config.list_physical_devices())

# 1. Improved model definition using Input layer
from tensorflow.keras import layers, models

model = models.Sequential(
    [
        layers.Input(shape=(20,)),  # Explicit input layer
        layers.Dense(32, activation="relu"),  # Hidden layer
        layers.Dense(2, activation="softmax"),  # Output layer
    ]
)

# 2. Compile the model with recommended settings
model.compile(
    optimizer="adam", loss="sparse_categorical_crossentropy", metrics=
)

# 3. Model Summary for verification
model.summary()

# 4. Dummy data for testing (10 samples, 20 features)
```

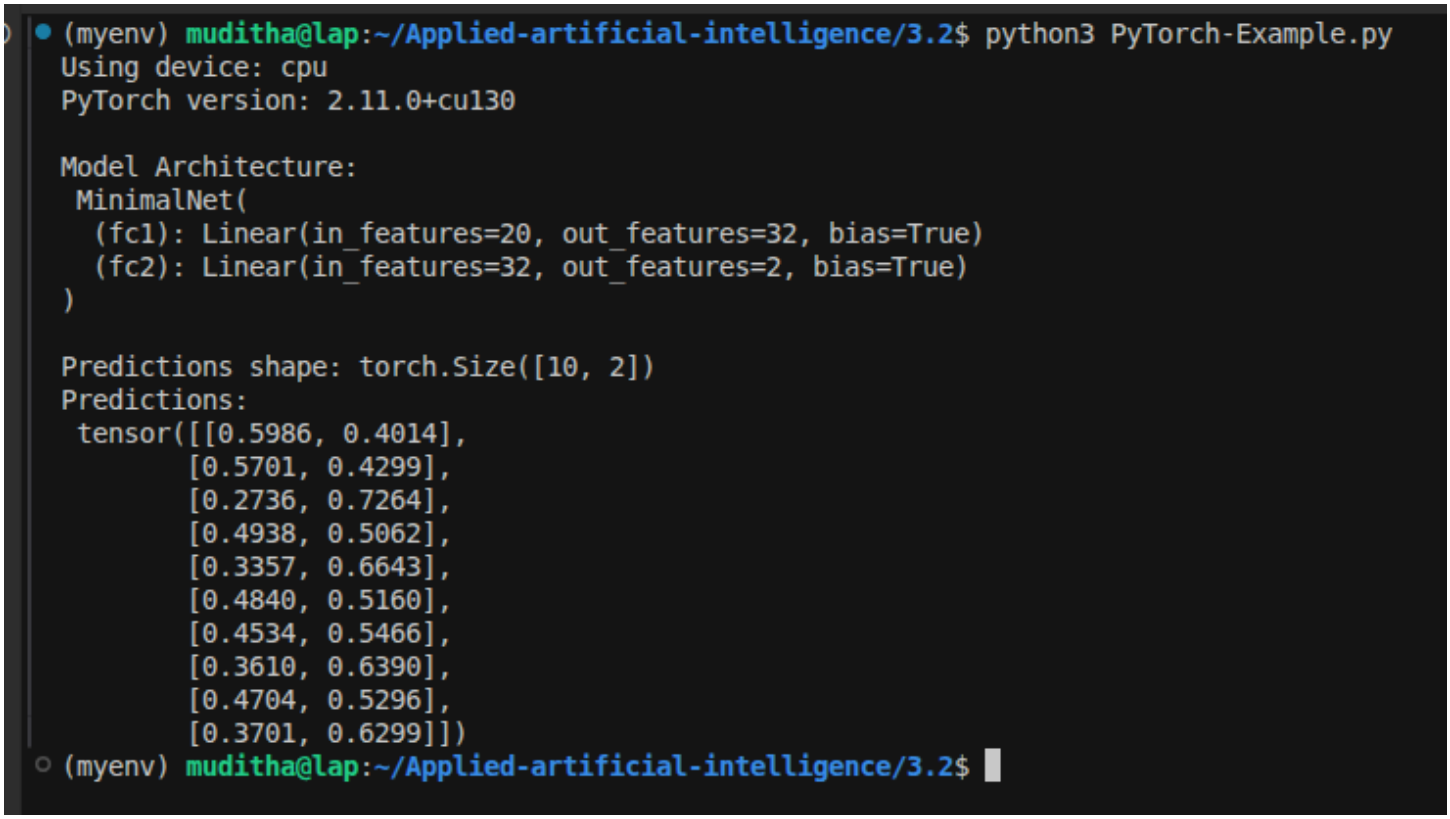
```
data = np.random.rand(10, 20)

# 5. Execute prediction
predictions = model.predict(data)
print("\nPredictions shape:", predictions.shape)
print("Predictions:\n", predictions)
```

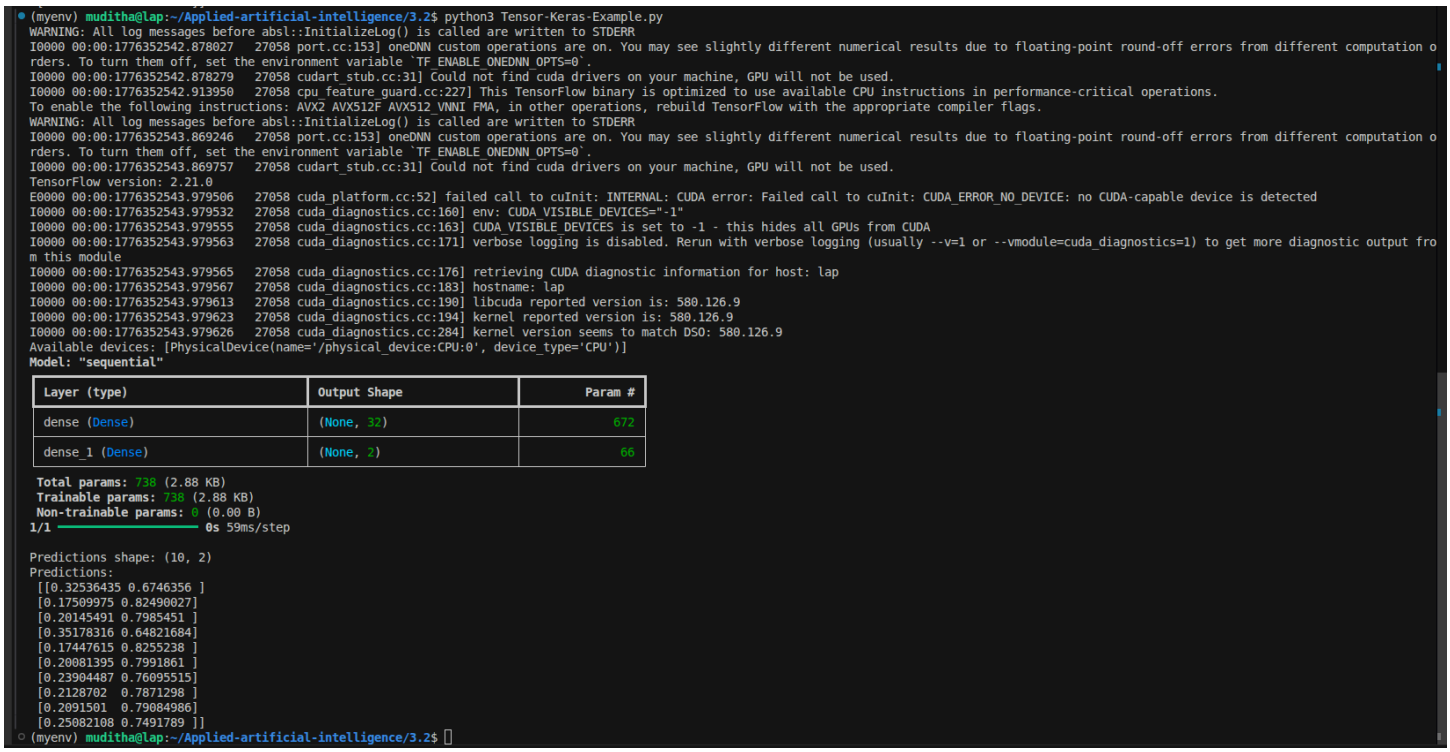
# PyTorch Example

```
import torch
import torch.nn as nn
import torch.nn.functional as F

# Force PyTorch to use CPU
device = torch.device("cpu")
print(f"Using device: {device}")
print("PyTorch version:", torch.__version__)


# 1. Define a simple model architecture
class MinimalNet(nn.Module):
    def __init__(self):
        super(MinimalNet, self).__init__()
        self.fc1 = nn.Linear(20, 32)  # Input layer to hidden
        self.fc2 = nn.Linear(32, 2)  # Hidden to output

    def forward(self, x):
        x = F.relu(self.fc1(x))
        x = self.fc2(x)
        return F.softmax(x, dim=1)


# 2. Initialize the model and move to CPU
model = MinimalNet().to(device)
print("\nModel Architecture:\n", model)

# 3. Generate dummy data (10 samples, 20 features)
# PyTorch requires data to be in Tensor format
data = torch.randn(10, 20).to(device)

# 4. Run prediction (forward pass)
model.eval()  # Set to evaluation mode
with torch.no_grad():
    predictions = model(data)
```

```
print("\nPredictions shape:", predictions.shape)
print("Predictions:\n", predictions)
```

## Console Output Screenshot:




## Test Results Summary:

- ✅ Model creation: Success

- ✅ Predictions generated: Success
- ✅ GPU utilized: No
- ✅ No errors: No

# 5. Challenges & Solutions

## Challenge 1

**Problem:** Upon running the initial test scripts, the system encountered multiple CUDA_ERROR_NO_DEVICE errors and warnings indicating that CUDA drivers could not be found. Because the local machine uses an older VGA/GPU that does not support the required CUDA versions for TensorFlow 2.21.0, the framework was unable to initialize hardware acceleration.

**Solution:** The issue was resolved by forcing the deep learning frameworks to operate exclusively on the CPU. For TensorFlow, this was achieved by setting the environment variable os.environ['CUDA_VISIBLE_DEVICES'] = '-1', which effectively hides any incompatible GPU from the library and prevents initialization errors. Additionally, the code was updated to use a dedicated layers.Input object to resolve UserWarnings regarding the model's architecture definition style. This ensured a stable, reproducible environment suitable for completing the assignment without specialized hardware.

# 7. Environment Documentation

## requirements.txt

```
absl-py==2.4.0
astunparse==1.6.3
certifi==2026.2.25
charset-normalizer==3.4.7
cuda-bindings==13.2.0
cuda-pathfinder==1.5.3
cuda-toolkit==13.0.2
filelock==3.25.2
flatbuffers==25.12.19
fsspec==2026.3.0
gast==0.7.0
google-pasta==0.2.0
grpcio==1.80.0
h5py==3.14.0
idna==3.11
Jinja2==3.1.6
keras==3.12.1
libclang==18.1.1
markdown-it-py==4.0.0
MarkupSafe==3.0.3
mdurl==0.1.2
ml_dtypes==0.5.4
mpmath==1.3.0
namex==0.1.0
networkx==3.4.2
numpy==2.2.6
nvidia-cublas==13.1.0.3
nvidia-cuda-cupti==13.0.85
nvidia-cuda-nvrtc==13.0.88
nvidia-cuda-runtime==13.0.96
nvidia-cudnn-cu13==9.19.0.56
nvidia-cufft==12.0.0.61
nvidia-cufile==1.15.1.6
```

```
nvidia-curand==10.4.0.35
nvidia-cusolver==12.0.4.66
nvidia-cusparse==12.6.3.3
nvidia-cusparselt-cu13==0.8.0
nvidia-nccl-cu13==2.28.9
nvidia-nvjitlink==13.0.88
nvidia-nvshmem-cu13==3.4.5
nvidia-nvtx==13.0.85
opt_einsum==3.4.0
optree==0.19.0
packaging==26.1
pillow==12.2.0
protobuf==7.34.1
Pygments==2.20.0
requests==2.33.1
rich==15.0.0
six==1.17.0
sympy==1.14.0
tensorflow==2.21.0
termcolor==3.3.0
torch==2.11.0
torchaudio==2.11.0
torchvision==0.26.0
triton==3.6.0
typing_extensions==4.15.0
urllib3==2.6.3
wrapt==2.1.2
```

### How to reproduce this environment:

```
# Step-by-step commands
1. python -m venv myenv
2. source myenv/bin/activate  # On Windows: myenv\Scripts\activate
3. pip install -r requirements.txt
```

# Appendix: Code Repository

**GitHub/Repository Link:** https://github.com/Muditha-Kumara/Applied-artificial-intelligence/tree/main/3.2

**Git Commit:** 88ac712722c30c30f8a4d07873bc983a0b28acb1